$\Delta^2 \log\tau$ under F_1, reduced

Cluster anatomy of the Uvarov τ-field, rank-one locality, and why F_1 does not close F_ε

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Abstract

Round 392 reduced F_ε to a bound on $\Delta^2 \log\tau$ for F_1-bounded ν-runs (max run ≤ 2, r382, MAIN 85/85), with the box $|e^{\Delta^2 \log\tau} - 1| \le \frac{1}{4}$ equivalent to $|\gamma - \frac{1}{4}| \le \frac{1}{16}$ on discrete Bernstein–Szegő, and the last-12 jump of γ equal to $|\Delta(\Delta^2 \log\tau)|$. This note splits Ξ under F_1 into singletons and pairs, writes τ as an explicit $1 \times 1/2 \times 2$ product times a coupling determinant κ, and proves that $\Delta^2 \log\tau$ is the first difference of a rank-one increment (matrix-determinant lemma) — the cluster-count does not enter linearly. What is proved: the decomposition identity; rank-one locality; coupling is load-bearing ($\sim 31\%$ of $|\Delta^2|$ on FRAME-A $w = 9$); a consecutive run of length 3 is a 3×3 block that leaves the pair bound (jump 0.4619, out even at the legal V_3' relaxation $\mathrm{JUMP}' = 0.45$); two-period AP has $\Delta^2 = 0$. What is *not* proved: the bound itself. F_1 is necessary and not sufficient — a random F_1 half-filling on the cosine grid leaves every legal jump constant (h$= 80$, jump 1.18); the cluster L^1 triangle ($0.436 > \log\frac{5}{4}$) does not close the box; raising JUMP to 0.45 covers the isolated-pair toy (0.4183) and the kz 55 census row (0.3942) and does not make either a theorem of F_1. No RH claim.

Status. Lemmas 1–3 are proved (finite identities). Lemma 4 is a machine identity. Propositions 1–2 name the reduction. No RH claim. No L^* claim. No G_ε claim.

1 Recollection

Occupied Fejér is node deletion of the ν-support Ξ from the discrete Bernstein–Szegő cosine grid [1, 2]. Writing $\tau_n = \det(I - K_n[\Xi]W)$ with $\tau_0 = 1$,

$$\gamma_n(\mu) = \gamma_n(\sigma) \cdot \frac{\tau_{n+1}\,\tau_{n-1}}{\tau_n^2}.$$

On $\gamma_n(\sigma) = \frac{1}{4}$ the box $|\gamma - \frac{1}{4}| \le \frac{1}{16}$ is $|e^{\Delta^2 \log\tau} - 1| \le \frac{1}{4}$, and the last-12 jump of γ is $|\Delta(\Delta^2 \log\tau)|$. F_1 (r382) is max consecutive ν-run ≤ 2 on the cosine grid, MAIN 85/85. V_3'/A_{15} first fails at coherent last-12 scale 2.0; scale 1.5 stays inside [4]. Raising JUMP from $\frac{2}{5}$ to 0.45 is therefore legal air; $\log\frac{5}{3} \approx 0.511$ would make the jump redundant with the box and is not used.

2 Cluster anatomy

Under F_1, Ξ is a disjoint union of singletons and adjacent pairs. Write K_n for the CD Gram of σ on Ξ.

Lemma 1 (1×1 and 2×2)**.** *A singleton ξ of mass w contributes*

$$\tau_n^{(\xi)} = 1 - w\, K_n(\xi, \xi)$$

(Sherman–Morrison). An adjacent pair (ξ_1, ξ_2) of masses (w_1, w_2) contributes the 2×2 determinant

$$\tau_n^{(\xi_1 \xi_2)} = (1 - w_1 K_{11})(1 - w_2 K_{22}) - K_{12}^2 w_1 w_2.$$

Exact over $\mathbb{Q}$ on a five-atom toy (singleton $1/5$; adjacent pair $2/15$).

Lemma 2 (decomposition)**.** *Writing $\tau_n^{\mathrm{diag}} = \prod_c \tau_n^{(c)}$ over the F_1 clusters and $\kappa_n = \tau_n / \tau_n^{\mathrm{diag}}$,*

$$\Delta^2 \log \tau = \sum_c \Delta^2 \log \tau^{(c)} + \Delta^2 \log \kappa.$$

The identity holds at $3 \cdot 10^{-14}$ on FRAME-A $w = 9$ (last 20 degrees). Two far singletons give $\kappa = 5/9$; two neighbours give a strictly larger $|\log \kappa|$ (CD decay of K_{12}).

On $w = 9$: $n_\nu = 104$ nodes split as 102 clusters (100 singletons, 2 pairs), F_1 true. Last-12 $\max|\Delta^2 \log \tau| = 0.1553$, $|e^{\Delta^2} - 1|_{\max} = 0.1438 < \frac{1}{4}$, $|\Delta(\Delta^2)|_{\max} = 0.2379$ equal to the occupied-Fejér jump. The diagonal piece is 0.1462; the coupling piece is 0.0490 ($\sim 31\%$ of the full last-12). Coupling is load-bearing: omitting $\Delta^2 \log \kappa$ is a named kill (error 0.049).

3 Rank-one locality

The cluster count scales with N. A triangle bound $\sum_c |\Delta^2 \log \tau^{(c)}|$ is therefore the wrong object unless Δ^2 cancels the volume.

Lemma 3 (rank-one increment)**.** *$K_{n+1} = K_n + \pi_n \pi_n^\top$. The matrix-determinant lemma gives*

$$\rho_n := \frac{\tau_{n+1}}{\tau_n} = 1 - (W \pi_n)^\top (I - K_n W)^{-1} \pi_n,$$

hence

$$\Delta^2 \log \tau_n = \log \rho_{n+1} - \log \rho_n.$$

Exact at $9 \cdot 10^{-15}$ on $w = 9$. Adjacent degrees share every cluster contribution except the rank-one update at n: Δ^2 reduces the volume sum to a local increment in the degree window. That is the structural statement of this round. It does not *by itself bound $|\log \rho_{n+1}/\rho_n|$.*

On $w = 9$ the last-12 L^1 of the cluster pieces is 0.436, against $|\sum| = 0.066$ (cancellation 0.85) and against $\log \frac{5}{4} \approx 0.223$. The triangle does not close the box. Per-cluster last-12 maxabs has median 0.0072 and max 0.0125; $n_{\mathrm{cl}} \times \mathrm{med} \approx 0.73$ is five times the coherent sum.

4 Adversaries

Lemma 4 ($w = 9$ pins)**.** *FRAME-A $w = 9$ satisfies F_1 with the cluster split of Lemma 2; last-12 occupied Fejér lies in the box at jump 0.2379. Machine identity against the Uvarov formula at $9 \cdot 10^{-15}$ is r392.*

Proposition 1 (F_1 necessary, not sufficient)**.** *A consecutive ν-run of length 3 on `full_grid`(40), last 12 of $n \le 60$, has jump 0.4619 and $|\gamma - \frac{1}{4}| = 0.0672 > \frac{1}{16}$: out at JUMP $= \frac{2}{5}$ and at JUMP$' = 0.45$. The 3×3 cluster Δ^2 on a height-12 toy exceeds the pair 2×2 ($|e^{\Delta^2} - 1| = 0.630 > 0.25$). F_1 is therefore necessary for the bound. It is not sufficient. Periodic F_1 tilings*

(1100*,* 1010*,* 110*) stay in at the same window; a random* F_1 *half-filling on* `full_grid`(80) *at* $n = 60$ *has jump* 1.179 *and leaves* JUMP$'$*. The isolated central pair (F1-legal, not half-filled) has jump* 0.4183*: out at* $\frac{2}{5}$*, in at* 0.45*, inside the* $1/16$ *box. Scramble occupation seed* 3 *has* max *run* 6 *(*12 *runs of length* ≥ 3*),* F_1 *false, occupied jump* 0.4438*: a cluster break, not a coupling remainder. Core-*42 *occupied last-12:* $0/42$ *out at* $\frac{2}{5}$*, max jump* 0.3942 *at kz* 55 *— census of the* L^**-folded class, not a corollary of* F_1*.*

Two-period AP deletion makes every cluster contribution identical across n, so $\Delta^2 \log \tau = 0$, consistent with the r392 non-kill (still Bernstein–Szegő). Wrong Sherman–Morrison sign on the cluster formula is a mutant (last-12 error 0.199).

5 What remains

Proposition 2 (reduction)**.** *The* τ*-field under* F_1 *is SATZ as algebra (Lemmas 1–3).* F_ε *does* not *follow from* F_1 *+ half-filling + mesh separation: random* F_1 *half-fillings leave every legal jump constant, and the cluster triangle does not close. Raising JUMP to* 0.45 *(legal* V_3' *air) covers the isolated-pair toy and the kz* 55 *margin* 0.0058*, and still leaves* F_ε *a census of the* L^**-folded occupation, not a theorem of* F_1*. The remaining named rest is* L^**-occupation regularity — the gap between* F_1*-legal cosine masks and the folded half-filling that actually occurs — sibling to the Sign-Schur lemma of [1] for Assist. The chain*

$$F_\varepsilon \iff \big(\Delta^2 \log \tau \ \textit{under } F_1\big) \iff \big(L^*\textit{-regular } F_1 \ \textit{occupation}\big)$$

is the honest reduction. W_ε *is untouched. No* m_4*. No RH claim.*

6 Machine checks

Numbered lemmas: `verify_tau_field.py` (same directory). Standalone: $1 \times 1/2 \times 2$ over $\mathbb{Q}$; κ vs gap; $\tau_0 = 1$; JUMP air. Construction: $w = 9$ F_1 split, rank-one, decomposition, coupling, AP non-kill, run-3 kill, scramble F_1-break, coupling mutant. Discovery census: `tau_field_probe.py` (`--smoke` / full), core-42, random-F_1 counterexample. Exit line: `TAU FIELD VERIFIED`.

Claim boundary

Finite identities for the Uvarov τ-field of a positive atomic measure on a cosine grid under F_1-bounded runs, a named reduction of F_ε to L^*-occupation regularity, and named clustered / random-F_1 / scramble breaks. No statement about zeros of $\zeta(s)$, in either direction. No RH claim.

References

[1] S. Hamann, *The deletion transform, reduced*, standalone note, August 2026, `rh/problem/deletion_transform.tex`.

[2] S. Hamann, *The* G_ε^μ*-lemma, reduced*, standalone note, August 2026, `rh/problem/g_eps_mu.tex`.

[3] S. Hamann, *The pivot-entry lemma*, standalone note, August 2026, `rh/problem/pivot_entry_lemma.tex`.

[4] S. Hamann, *The V_3'-lemma, reduced*, standalone note, August 2026, `rh/problem/v3prime_proof.tex`.